

EXP NO:6A

DATA VISUALIZATION USING POWER BI (RETAIL SALES ANALYSIS)

AIM:

To import, transform, analyze and visualize Baseball Databank data using Power BI Desktop and create an interactive dashboard and reports for baseball performance analysis.

ALGORITHM:

1. Start Power BI Desktop.
2. Import the Baseball Databank table using Get Data.
3. Check columns and clean data using Power Query.
4. Remove duplicate records and handle missing values.
5. Create calculated fields/measures.
6. Create bar, pie/donut, line and card visuals.
7. Add titles, labels, filters and slicers.
8. Arrange visuals into an interactive dashboard.
9. Analyze important patterns and save the Power BI report

OUTPUT:

```
In [1]: from pathlib import Path

def find_data(filename):
    candidates = [
        Path(filename),
        Path("data") / filename,
        Path("../data") / filename,
        Path("../..data") / filename,
        Path("/content") / filename,
    ]
    for p in candidates:
        if p.exists():
            return p
    raise FileNotFoundError(f"{filename} not found. Upload the Baseball Databank CSV or keep it in the repository data folder.")

import pandas as pd
import numpy as np
import matplotlib.pyplot as plt

file_path = find_data("Batting.csv")
df = pd.read_csv(file_path)
print("Dataset shape:", df.shape)
print("Columns:", list(df.columns))
print("Duplicate rows:", df.duplicated().sum())

Dataset shape: (101332, 22)
Columns: ['playerID', 'yearID', 'stint', 'teamID', 'lgID', 'G', 'AB', 'R', 'H', '2B', '3B', 'HR', 'RBI', 'SB', 'CS', 'BB', 'SO', 'IBB', 'HBP', 'SH', 'SF', 'GIDP']
Duplicate rows: 0
```

```
In [2]: print("Missing values:")
        print(df.isnull().sum())
```

```
Missing values:
playerID      0
yearID        0
stint         0
teamID        0
lgID          737
G             0
AB            5149
R             5149
H             5149
2B            5149
3B            5149
HR            5149
RBI           5573
SB            6449
CS            28603
BB            5149
SO            12987
IBB           41712
HBP           7959
SH            11487
SF            41181
GIDP          31257
dtype: int64
```

```
In [3]: clean = df.copy()

        for col in clean.select_dtypes(include=np.number).columns:
            clean[col] = clean[col].fillna(0)

        for col in clean.select_dtypes(include=["object"]).columns:
            clean[col] = clean[col].fillna("Unknown")

        print("Prepared shape:", clean.shape)
        print("Remaining missing values:", int(clean.isnull().sum().sum()))
```

```
Prepared shape: (101332, 22)
Remaining missing values: 0
```

```
In [4]: total_runs = clean["R"].sum()
        total_players = clean["playerID"].nunique()
        total_seasons = clean["yearID"].nunique()
        total_records = len(clean)

        print("Total Runs =", total_runs)
        print("Total Players =", total_players)
        print("Total Seasons =", total_seasons)
        print("Total Player-Season Records =", total_records)
```

```
Total Runs = 1912795.0
Total Players = 18659
Total Seasons = 145
Total Player-Season Records = 101332
```

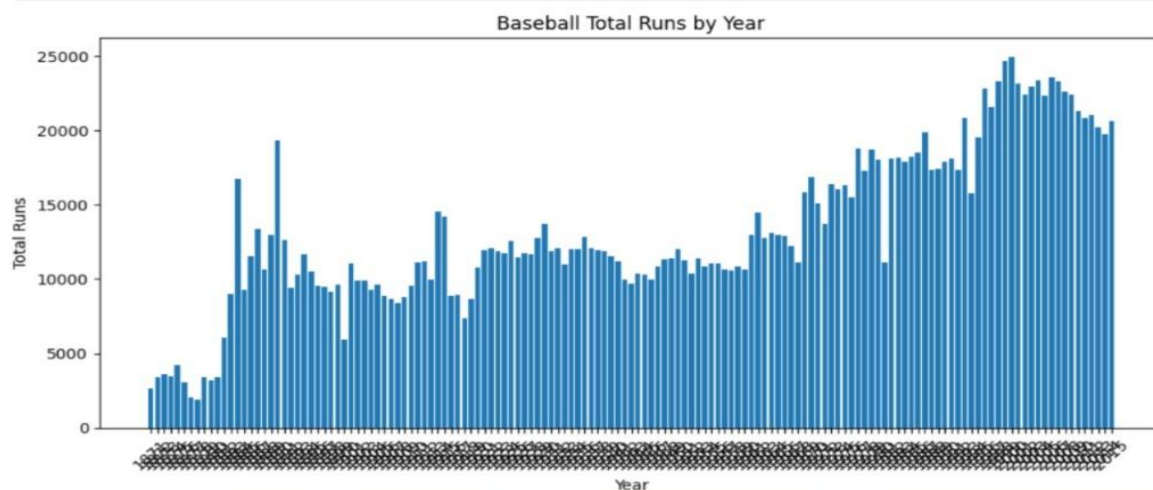
```
In [5]: clean["total_runs"] = clean["R"]
        season_runs = clean.groupby("yearID", as_index=False)["total_runs"].sum()
        print(season_runs.to_string(index=False))
```

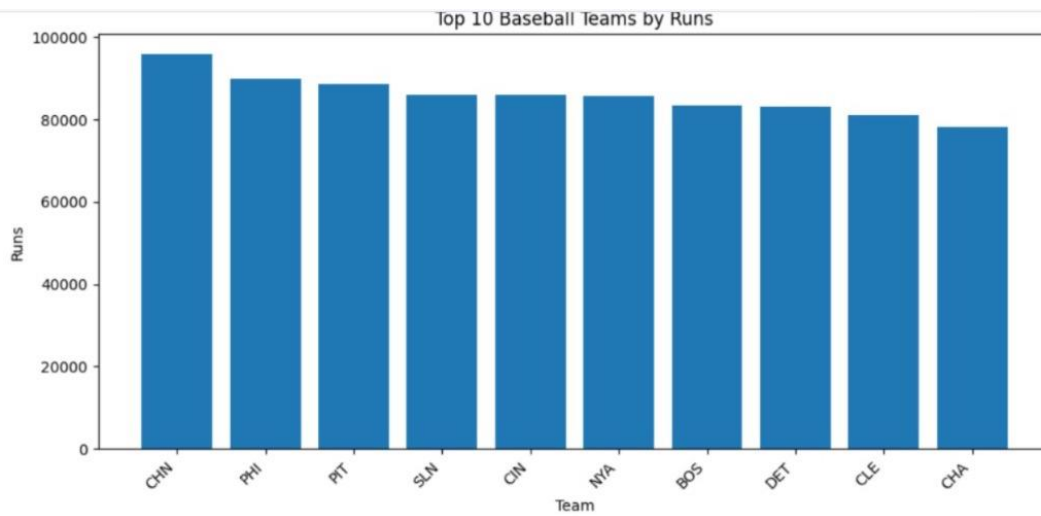
```
yearID  total_runs
1871    2659.0
1872    3390.0
1873    3580.0
1874    3470.0
1875    4234.0
1876    3066.0
1877    2040.0
1878    1904.0
1879    3409.0
1880    3191.0
1881    3426.0
1882    6095.0
1883    9030.0
1884    16742.0
1885    9292.0
1886    11512.0
1887    13414.0
1888    10628.0
1889    12986.0
1890    19330.0
1891    12667.0
1892    9388.0
1893    10315.0
1894    11678.0
1895    10482.0
1896    9555.0
1897    9520.0
1898    9129.0
1899    9647.0
1900    5932.0
1901    11073.0
1902    9879.0
1903    9888.0
```

```
In [7]: team_runs = clean.groupby("teamID")["R"].sum().sort_values(ascending=False).head(10)
print(team_runs.to_string())
plt.figure(figsize=(10,5))
plt.bar(team_runs.index, team_runs.values)
plt.title("Top 10 Baseball Teams by Runs")
plt.xlabel("Team")
plt.ylabel("Runs")
plt.xticks(rotation=45, ha="right")
plt.tight_layout()
plt.savefig("Experiment_6A_Baseball_Team_Bar.png", dpi=150, bbox_inches="tight")
plt.show()
```

```
teamID
CHN    96003.0
PHI    89959.0
PIT    88658.0
SLN    86003.0
CIN    85970.0
NYA    85715.0
BOS    83309.0
DET    83222.0
CLE    81116.0
CHA    78221.0
```

```
plt.figure(figsize=(10,5))
plt.bar(season_runs["yearID"].astype(str), season_runs["total_runs"])
plt.title("Baseball Total Runs by Year")
plt.xlabel("Year")
plt.ylabel("Total Runs")
plt.xticks(rotation=45)
plt.tight_layout()
plt.savefig("Experiment_6A_Baseball_Season_Line.png", dpi=150, bbox_inches="tight")
plt.show()
```



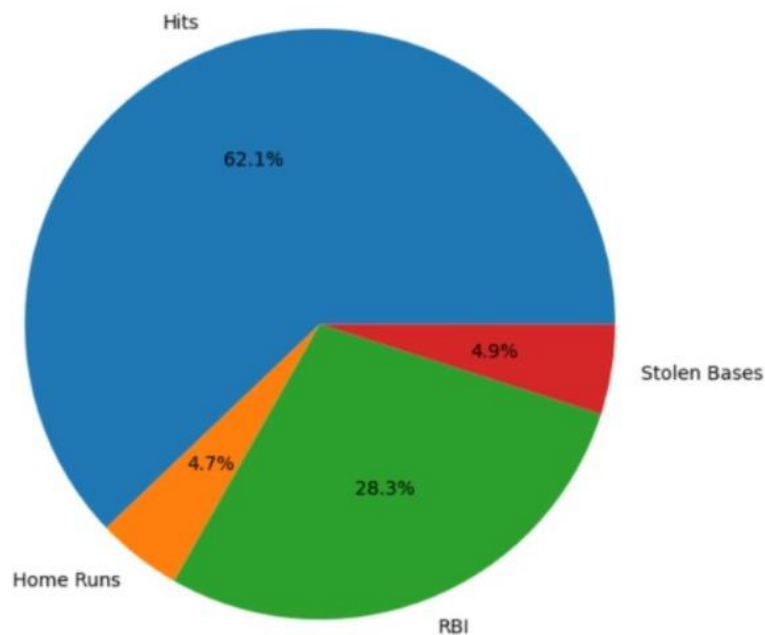


```
In [8]: components = pd.Series({"Hits": clean["H"].sum(), "Home Runs": clean["HR"].sum(), "RBI": clean["RBI"].sum(), "Stolen Bases": clean["SB"].sum()})
print(components.to_string())
plt.figure(figsize=(7,7))
plt.pie(components.values, labels=components.index, autopct="%1.1f%%")
plt.title("Selected Baseball Batting Totals")
plt.savefig("Experiment_6A_Baseball_Metric_Pie.png", dpi=150, bbox_inches="tight")
plt.show()
```

```
In [8]: components = pd.Series({"Hits": clean["H"].sum(), "Home Runs": clean["HR"].sum(), "RBI": clean["RBI"].sum(), "Stolen Bases": clean["SB"].sum()})
print(components.to_string())
plt.figure(figsize=(7,7))
plt.pie(components.values, labels=components.index, autopct="%1.1f%%")
plt.title("Selected Baseball Batting Totals")
plt.savefig("Experiment_6A_Baseball_Metric_Pie.png", dpi=150, bbox_inches="tight")
plt.show()
```

```
Hits      3776303.0
Home Runs  283673.0
RBI        1720326.0
Stolen Bases 299658.0
```

Selected Baseball Batting Totals



```
In [9]: print("DAX Measure examples for Power BI:")
        print("Total Runs = SUM(Batting[R])")
        print("Total Players = DISTINCTCOUNT(Batting[playerID])")
        print("Total Seasons = DISTINCTCOUNT(Batting[yearID])")
        print("Total Records = COUNTROWS(Batting)")
```

```
DAX Measure examples for Power BI:
Total Runs = SUM(Batting[R])
Total Players = DISTINCTCOUNT(Batting[playerID])
Total Seasons = DISTINCTCOUNT(Batting[yearID])
Total Records = COUNTROWS(Batting)
```

```
In [10]: print("Result: The Baseball Databank was successfully prepared for Power BI analysis.")
```

```
Result: The Baseball Databank was successfully prepared for Power BI analysis.
```

RESULT:

Thus, the Baseball Databank was successfully prepared and analyzed for Power BI, with KPIs and visualizations for runs, players, seasons and team performance.